

1. GENERAL

- 1.1 TITLE: **Ceramic Processing**
- 1.2 COURSE NUMBER (if known): **DC.MCR 304.19**
- 1.3 CREDITS: **9 (L-T-P: 3-0-0)**
- 1.4 SEMESTER-OFFERED: **V Semester**

2. OBJECTIVE:

The objective of this course is to acquaint the student with the knowledge and significance of ceramic processing. The course aims to offer comprehensive insights into how the processing methodology notably influences their structure, properties, and performance. To develop an understanding of advanced ceramic processing techniques. To establish processing-structure-property relationships.

3. COURSE TOPICS:

Ceramic Processing: Solid state method for synthesis of Ceramics. Freeze drying, Spray drying. Co-precipitation method and Sol-Gel process. Chemical vapor deposition, mechanochemical synthesis, hydrothermal synthesis and plasma spraying. Combustion method for synthesis of ceramics. Synthesis of nano-ceramics by physical and chemical methods.

Sintering: Driving force, defects and defects chemistry, diffusion in solids, vapor pressure over curved surface. Theory of solid state and viscous sintering, scaling laws. Hot Pressing. Grain Growth: Normal and abnormal grain growth, Second phase particles, grain boundary phenomena. Liquid Phase Sintering. Stages in liquid phase sintering, Basic Mechanisms of liquid phase sintering.

4. READINGS

4.1 TEXTBOOKS:

1. M. N. Rahaman: Ceramic Processing and Sintering, CRC Press, 2003.
2. J. S. Reed: Introduction to the Principles of Ceramic Processing, Wiley, 1988

4.2 *REFERENCE BOOKS:

1. W. D. Kingery, H. K. Bowen, D. R. Uhlmann: Introduction to Ceramics, Wiley, 1976
2. M. W. Barsoum: Fundamentals of Ceramics, CRC Press, 2003
3. W. D. Callister Jr.: Materials Science and Engineering: An Introduction, Wiley, 2007